Nonlinear functions — Part 1 answer key

#	ID	Answer	Difficulty
1.1	b58ee74d-9a79-4ad3-b997-7fb216ab725f	B	Easy
1.2	1549b7a3-5fbf-46f8-8861-103e5d9edae0	C	Easy
1.3	43c1b9bd-91a3-4f33-8ec6-ca10f3f10196	6	Easy
1.4	fc448e0e-6144-425a-a9b3-2ff8ee674f1b	3	Easy
1.5	e0bc2b05-f917-4d64-b9ad-36fa20c0b430	C	Easy
1.6	58bfb590-016f-4bec-8d23-459d22e7482a	C	Easy
1.7	752821ca-8045-45d2-afe6-e57b8760f5f1	7	Easy
1.8	84e5072e-3d6b-4453-a0ee-23b7b7999691	A	Easy
1.9	e9d27a82-510c-4c48-a723-ee345529a8fb	B	Easy
1.10	3212a20d-5aee-41d8-8981-4e3ce65c7918	B	Easy
1.11	f827079b-37e0-4696-9873-897b36b24545	C	Easy
1.12	cecc8739-2d4c-4044-8624-baee7ca16ce6	A	Easy
1.13	f45f8ef3-75e8-4989-991d-8bcf481614f7	A	Easy
1.14	75b672c9-de51-448e-b5a4-f2bd63b15e93	A	Easy
1.15	29fc6f6a-c78f-4294-a4fc-98e4343406e7	C	Easy
1.16	e07cc332-1b7d-4d38-a472-8db2ff2d4e41	B	Easy
1.17	d6f230e9-a5fd-4506-837c-cde890c380bc	4	Easy
1.18	37ced69c-527a-4d31-a90e-19a862272659	B	Easy
1.19	d6c4b6c3-23fb-477d-aba3-9ed20b31fbb9	C	Easy
1.20	d73b26c3-ce67-4987-8411-3f6ae5fa6ca3	A	Easy
1.21	dc882084-e2b9-4e68-92b2-78a7a8f89acd	B	Easy
1.22	e599f748-6055-4c3c-83f8-5366aad721b3	B	Easy

1.23	a6fb9475-3880-4ae5-952c-4bbb83ee59af	12	Easy
1.24	591c966e-550b-4b07-97bb-f659afe623d8	40	Easy
1.25	4e837070-2e7d-4f56-993a-9d8d616e345b	B	Easy
1.26	73405bf8-4af6-4fad-9344-e38505f18416	A	Easy
1.27	d38233c5-51bc-4d5b-b405-07bfe9387efc	A	Easy
1.28	b0bf4cfd-4fbe-4424-acc0-d7a7f44468f7	C	Easy
1.29	0eaf53b0-0c3b-45a2-bf0d-c423f1999a0d	B	Easy
1.30	518e7cca-1f28-4828-8926-524f47cebb81	A	Easy
1.31	16aeb5f1-86d3-4fcc-b2ef-5ad34f8f79f5	C	Easy
1.32	af1af0e8-fb4f-4830-9348-cc9508783406	C	Easy
1.33	a31258b0-0750-420d-9cc1-2b4aad1e6dc2	B	Easy
1.34	3c84e10c-eecd-4a7b-b28c-a5de3d455977	D	Easy
1.35	56af9a6d-1c30-4ea5-a528-45b60f366d79	3	Easy
1.36	2c8d6c87-c4db-4555-8418-f31510d6ec98	31	Easy
1.37	05723-DC	A	Easy
1.38	d359151d-6c08-4338-b308-8a73c712373f	A	Easy
1.39	b0ead772-4a2d-45aa-95e5-204e1ccfdb59	A	Easy
1.40	032723-DC	C	Easy
1.41	3412b211-d5f8-400a-988e-f1dd95e0d2e3	C	Easy
1.42	a32cd185-f66c-43e0-b4ca-8c66da898744	-3	Easy
1.43	e647a4e3-60a1-4bf3-8ceb-4b40a8d430f1	D	Easy
1.44	5943941c-677e-4c02-9da5-eddac62bf566	D	Easy
1.45	7af6ef83-b534-4900-b63b-a74eee6196f9	B	Easy
1.46	027806-DC	B	Easy
1.47	08dcb1a6-8ea6-4767-9d5b-83bca79b58fa	23	Easy
1.48	cd3ef442-8790-4484-ae9f-8b639799e10a	D	Easy

1.49	6f8b1a0e-8c22-4208-8038-796a6de480d4	3	Easy
1.50	f7def592-aac8-4d10-bb76-680518e870b2	77	Easy
1.51	83b9a6a0-309d-476f-bd88-5134eb899234	B	Easy
1.52	ff1de39f-4c5b-4dcb-8710-09663efcf4f5	7	Easy
1.53	2d504a02-4e7f-409c-a455-32753236fe82	B	Easy
1.54	edb987d4-772f-4ea6-8b75-8cd93a8fc5c8	A	Easy
1.55	021441-DC	D	Easy
1.56	5d453a5b-d7a9-40ac-92b4-8a1f6744f99e	A	Easy
1.57	87aede6d-89e4-4d41-9c3c-af1ae260cc3a	3	Easy
1.58	6343862e-4b55-4888-a763-6c005c8707e0	D	Easy
1.59	ec7fe4db-233b-4c49-9a70-2816bda45ba9	A	Easy
1.60	52978a8b-cb33-43f8-a130-45f92771e3d0	D	Easy
1.61	025654-DC	C	Easy
1.62	021520-DC	C	Easy
1.63	2deb518b-5f4a-45ec-828f-20051b7eb7aa	D	Easy
1.64	1cfbd034-f610-4c8e-8e6f-88944150668f	A	Easy
1.65	2d05a911-7298-4d94-ad93-ec0fefdedb1d	C	Easy
1.66	ff0ab105-aa85-4cb3-a07f-f00a7c55c4e3	15000	Medium
1.67	62b8544c-e4cc-4953-8687-0729c0554f61	A	Medium
1.68	00d08c52-5d90-46ea-83d3-f3cc7ab08588	C	Medium
1.69	40adf1b6-46fb-41b3-a10c-d6c1ab8c28c8	3600	Medium
1.70	b5fc9e88-ea7b-46f4-af78-c536a42d0baf	A	Medium
1.71	62aee8c9-96f8-4f03-aad4-f068bd524749	B	Medium
1.72	35058beb-41fe-4d27-9257-454cd92cb3f3	D	Medium
1.73	faf32f9b-26f8-4ad5-8215-b6f52b1118f5	C	Medium
1.74	10571-DC	A	Medium

1.75	032669-DC	C	Medium
1.76	91bb6f2f-9d21-4a10-add7-5e0ad39aece8	B	Medium
1.77	bcebeee7-1522-4888-bc6b-06a4c5b1dae3	C	Medium
1.78	d1baa89a-a42c-46e1-a485-7574f59a8753	D	Medium
1.79	c4bea442-d7d0-404b-a78a-6881543f941b	B	Medium
1.80	af32328c-4e46-4a53-b9c0-f627a4d608f9	D	Medium
1.81	783877b4-b840-4260-8d01-4234651d93c4	14, -5, -4	Medium
1.82	2b50d060-8469-4554-937e-da992751dc98	B	Medium
1.83	076cf150-d233-4c88-b49e-49a87889ba92	A	Medium
1.84	e3e1bd01-4a4e-4c9c-8e91-28a5165667b6	-6, 4	Medium
1.85	e78c85a7-41e7-485b-b784-99727df47183	C	Medium
1.86	b790c80c-3eb0-4ef6-8101-d653f1487b32	B	Medium

Rationales

▼ 1.1 — B

Choice B is correct. The *y*-intercept of a graph in the *xy*-plane is the point where $x = 0$. For the given equation, the *y*-intercept of the graph in the *xy*-plane can be found by substituting 0 for x in the equation, which yields $y = -\frac{1}{4}(0)^2 + 20 + 29$, or $y = 29$. Therefore, the *y*-intercept of the graph is $(0, 29)$. It's given that y is the estimated number of scheduled deliveries x months after the end of May 2012. Therefore, $x = 0$ represents 0 months after the end of May 2012, or the end of May 2012. Thus, the best interpretation of the *y*-intercept of the graph of this equation in the *xy*-plane is that at the end of May 2012, the estimated number of scheduled deliveries was 29.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 1.2 — C

Choice C is correct. Each point x, y on the graph represents an elapsed time x, in hours, and the corresponding ocean water level y, in feet, at a certain location based on the model. The graph shown passes through the points $0, 0$, $3, -12$, and $6, 0$. Thus, the table in choice C gives the values of x and their corresponding values of y based on the model.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.3 — 6

The correct answer is 6. An x-intercept of a graph is a point on the graph where it intersects the x-axis, or where the value of y is 0. The graph shown intersects the x-axis at the point $6, 0$. Therefore, the x-coordinate of the x-intercept of the graph shown is 6.

▼ 1.4 — 3

The correct answer is 3. It's given that the y-intercept of the graph shown is $0, y$. The graph passes through the point $0, 3$. Therefore, the value of y is 3.

▼ 1.5 — C

Choice C is correct. It's given that $fx = 8\sqrt{x}$. Substituting 48 for fx in this equation yields $48 = 8\sqrt{x}$. Dividing both sides of this equation by 8 yields $6 = \sqrt{x}$. This can be rewritten as $\sqrt{x} = 6$. Squaring both sides of this equation yields $x = 36$. Therefore, the value of x for which $fx = 48$ is 36.

Choice A is incorrect. If $x = 6$, $fx = 8\sqrt{6}$, not 48.

Choice B is incorrect. If $x = 8$, $fx = 8\sqrt{8}$, not 48.

Choice D is incorrect. If $x = 64$, $fx = 8\sqrt{64}$, which is equivalent to 64, not 48.

▼ 1.6 — C

Choice C is correct. It's given that x represents years after 2010. Therefore, 2010 is represented by $x = 0$. On the model shown, the point with an x-coordinate of 0 has a y-coordinate of $20{,}000$. Thus, the model estimates that in 2010, the city had $20{,}000$ residents.

Choice A is incorrect. This is the value of x that represents the year 2010.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is approximately the number of residents the model estimates the city had in 2014, not 2010.

▼ 1.7 — 7

The correct answer is 7. It's given that the parabola intersects the y-axis at the point x, y. The graph shows that the parabola intersects the y-axis at the point $0, 7$. Therefore, the value of y is 7.

▼ 1.8 — A

Choice A is correct. Substituting 2 for x in the given equation yields $k(2) = 2^3 + 110$, which is equivalent to $k(2) = 8 + 110$, or $k(2) = 118$. Therefore, the value of $k(x)$ when $x = 2$ is 118.

Choice B is incorrect. This is the value of $k(x)$ when $x = 2$ if $k(x) = 3x + 110$, not $k(x) = x^3 + 110$.

Choice C is incorrect. This is the value of $k(x)$ when $x = 2$ if $k(x) = x + 3 + 110$, not $k(x) = x^3 + 110$.

Choice D is incorrect. This is the value of $k(x)$ when $x = 0$, not $x = 2$.

▼ 1.9 — B

Choice B is correct. The function P gives the estimated number of marine mammals in a certain area, where t is the number of years since a study began. Since the value of $P0$ is the value of Pt when $t = 0$, it follows that $P0 = 1{,}800$ means that the value of Pt is $1{,}800$ when $t = 0$. Since t is the number of years since the study began, it follows that $t = 0$ is 0 years since the study began, or when the study began. Therefore, the best interpretation of $P0 = 1{,}800$ in this context is the estimated number of marine mammals in the area was $1{,}800$ when the study began.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.10 — B

Choice B is correct. For a bacteria with an initial population of P cells and a doubling time of m minutes, the number of cells, fx, of the bacteria x minutes after the growth period begins can be given by the equation $fx = P2^{\frac{x}{m}}$. It's given that there are 830 cells of the bacteria when the growth period begins. Therefore, $P = 830$. It's also given that the number of cells of the bacteria doubles every 25 minutes during the growth period. Therefore, $m = 25$. Substituting 830 for P and 25 for m in the equation $fx = P2^{\frac{x}{m}}$ yields $fx = 8302^{\frac{x}{25}}$.

Choice A is incorrect. This function represents a situation where the number of cells decreases by half, rather than doubles, every 25 minutes.

Choice C is incorrect. This function represents a situation where the number of cells decreases by two-thirds, rather than doubles, every 25 minutes.

Choice D is incorrect. This function represents a situation where the number of cells triples, rather than doubles, every 25 minutes.

▼ 1.11 — C

Choice C is correct. It's given that function g is defined by $g(x) = \sqrt{x} + 300$. Substituting 81 for x in function g yields $g(81) = \sqrt{81} + 300$, which is equivalent to $g(81) = 9 + 300$, or $g(81) = 309$. Therefore, the value of $g(x)$ when $x = 81$ is 309.

Choice A is incorrect. This is the value of $\sqrt{81}$, not $\sqrt{81} + 300$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $81 + 300$, not $\sqrt{81} + 300$.

▼ 1.12 — A

Choice A is correct. It's given that $gx = x^2 + 9$. Substituting 25 for gx in this equation yields $25 = x^2 + 9$. Subtracting 9 from both sides of this equation yields $16 = x^2$. Taking the square root of each side of this equation yields $x = \pm 4$. It follows that $gx = 25$ when the value of x is 4 or -4. Only 4 is listed among the choices.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.13 — A

Choice A is correct. It's given that the kinetic energy, in joules, of an object with a mass of 9 kilograms traveling at a speed of v meters per second is given by the function K, where $Kv = \frac{9}{2}v^2$. It follows that in the equation $K34 = 5{,}202$, 34 is the value of v, or the speed of the object, in meters per second, and $5{,}202$ is the kinetic energy, in joules, of the object at that speed. Therefore, the best interpretation of $K34 = 5{,}202$ in this context is the object traveling at 34 meters per second has a kinetic energy of $5{,}202$ joules.

Choice B is incorrect. The object traveling at 340 meters per second has a kinetic energy of $520{,}200$ joules.

Choice C is incorrect. The object traveling at $5{,}202$ meters per second has a kinetic energy of $121{,}773{,}618$ joules.

Choice D is incorrect. The object traveling at $23{,}409$ meters per second has a kinetic energy of $2{,}465{,}915{,}764.5$ joules.

▼ 1.14 — A

Choice A is correct. It's given that the graph shows the height above ground, in meters, of a ball x seconds after the ball was launched upward from a platform. In the graph shown, the x-axis represents time, in seconds, and the y-axis represents the height of the ball above ground, in meters. It follows that for the marked point $1.0, 4.8$, 1.00 represents the time, in seconds, after the ball was launched upward from a platform and 4.80 represents the height of the ball above ground, in meters. Therefore, the best interpretation of the marked point $1.0, 4.8$ is 1.00 second after being launched, the ball's height above ground is 4.80 meters.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 1.15 — C

Choice C is correct. It's given that t represents the number of years since the account was opened. Therefore, $\frac{t}{10}$ represents the number of 10-year periods since the account was opened. Since the value of the account doubles during each of these 10-year periods, the value of the account can be found by multiplying the initial value by $\frac{t}{10}$ factors of 2. This is equivalent to $2^{\frac{t}{10}}$. It's given that the initial value of the account is $\$\,890$. Therefore, the value

of the account Mt, in dollars, t years after the account was opened can be represented by $Mt = 8902^{\frac{t}{10}}$.

Choice A is incorrect. This equation represents the value of an account if the value of the account halves, not doubles, every 10 years.

Choice B is incorrect. This equation represents the value of an account if the value of the account decreases by 90%, not doubles, every 2, not 10, years.

Choice D is incorrect. This equation represents the value of an account if the value of the account increases by a factor of 10, not doubles, every 2, not 10, years.

▼ 1.16 — B

Choice B is correct. The y-intercept of a graph is the point at which the graph intersects the y-axis. The graph shown intersects the y-axis at the point $(0, 8)$. Therefore, the y-intercept of the graph is $(0, 8)$. It's given that y is the height of the marble above the ground, in inches, and x is the number of seconds after the marble started moving. It follows that the marble's height was 8 inches above the ground 0 seconds after it started moving. Therefore, the best interpretation of the y-intercept of the graph is that the marble's height was 8 inches above the ground when it started moving.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 1.17 — 4

The correct answer is 4. The y-intercept of the graph of a function in the xy-plane is the point where the graph intersects the y-axis. The graph of the polynomial function shown intersects the y-axis at the point $(0, 4)$. It's given that the y-intercept of the graph is $(0, y)$. Therefore, the value of y is 4.

▼ 1.18 — B

Choice B is correct. The y-intercept of a graph in the xy-plane is the point x, y on the graph where $x = 0$. At $x = 0$, the corresponding value of y is -5. Therefore, the y-intercept of the graph shown is $0, -5$.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the y-intercept of a graph in the xy-plane that intersects the y-axis at $y = -4$, not $y = -5$.

Choice D is incorrect. This is the y-intercept of a graph in the xy-plane that intersects the y-axis at $y = 0$, not $y = -5$.

▼ 1.19 — C

Choice C is correct. The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. Since the value of $P8$ is the value of Pt when $t = 8$, it follows that "$P8$ is approximately equal to 32.91" means that the value of Pt is approximately equal to 32.91 when $t = 8$. Therefore, the best interpretation of the statement "$P8$ is approximately equal to 32.91" is that 8 years after 1984, the predicted population of this country was approximately 32.91 million.

Choice A is incorrect. In 1984, the predicted population of this country was 24.8 million, not approximately 8 million.

Choice B is incorrect. In 1984, the predicted population of this country was 24.8 million, not approximately 32.91 million.

Choice D is incorrect. 32.91 years after 1984, the predicted population of this country was $24.81.036^{32.91}$ million, or approximately 79.42 million, not approximately 8 million.

▼ 1.20 — A

Choice A is correct. It's given that the graph shown gives the estimated value y, in dollars, of a tablet as a function of the number of months since it was purchased, x. The y-intercept of a graph is the point at which the graph intersects the y-axis, or when x is 0. The graph shown intersects the y-axis at the point $0, 225$. It follows that 0 months after the tablet was purchased, or when the tablet was purchased, the estimated value of the tablet was 225 dollars. Therefore, the best interpretation of the y-intercept is that the estimated value of the tablet was $\$ 225$ when it was purchased.

Choice B is incorrect. The estimated value of the tablet 24 months after it was purchased was $\$ 50$, not $\$ 225$.

Choice C is incorrect. The estimated value of the tablet had decreased by $\$ 225 - \$ 50$, or $\$ 175$, not $\$ 225$, in the 24 months after it was purchased.

Choice D is incorrect and may result from conceptual errors.

▼ 1.21 — B

Choice B is correct. It's given that *y* represents the height, in feet, of the ball *x* seconds after it was launched. It's also given that during the first part of an experiment, a ball was launched from a 7-foot-tall platform. Therefore, the *y*-coordinate of the *y*-intercept of the given graph, 7, represents the platform height, in feet. During the second part of the experiment, the platform the ball was launched from was 2 feet shorter than the platform in the first part of the experiment. It follows that the height of the platform in the second part of the experiment was $7 - 2$ feet, or 5 feet. Therefore, the *y*-coordinate of the *y*-intercept of the graph representing the second part of the experiment must be 5. Only choice B satisfies this condition.

Choice A is incorrect. This could represent the graph if the ball were launched from a platform that was about 3 feet shorter rather than 2 feet shorter.

Choice C is incorrect. This could represent the graph if the ball were launched from a platform that was 2 feet taller rather than 2 feet shorter.

Choice D is incorrect. This could represent the graph if the ball were launched from a platform that was twice as tall rather than 2 feet shorter.

▼ 1.22 — B

Choice B is correct. It's given that for the function *h*, *hn* is the estimated maximum height, in feet, of the ball after it has hit the ground *n* times. It's also given that the function *h* estimates that the maximum height reached after each time the ball hits the ground is 85% of the maximum height reached after the previous time the ball hit the ground. It follows that *h* is a decreasing exponential function that can be written in the form $hn = a\frac{p}{100}^{n}$, where *a* is the initial height, in feet, the ball was dropped from and the function estimates that the maximum height reached after each time the ball hits the ground is *p*% of the maximum height reached after the previous time the ball hit the ground. It's given that the ball is dropped from an initial height of 22 feet. Therefore, $a = 22$. Since the function *h* estimates that the maximum height reached after each time the ball hits the ground is 85% of the maximum height reached after the previous time the ball hit the ground, $p = 85$. Substituting 22 for *a* and 85 for *p* in the equation $hn = a\frac{p}{100}^{n}$ yields $hn = 22\frac{85}{100}^{n}$, or $hn = 220.85^{n}$.

Choice A is incorrect. This function estimates that the maximum height reached after each time the ball hits the ground is 22%, not 85%, of the maximum height reached after the

previous time the ball hit the ground.

Choice C is incorrect. This function estimates that the ball is dropped from an initial height of 85 feet, not 22 feet, and that the maximum height reached after each time the ball hits the ground is 22%, not 85%, of the maximum height reached after the previous time the ball hit the ground.

Choice D is incorrect. This function estimates that the ball is dropped from an initial height of 85 feet, not 22 feet.

▼ 1.23 — 12

The correct answer is 12. The value of $f36$ is the value of fx when $x = 36$. Substituting 36 for x in the given equation yields $f36 = 6 + \sqrt{36}$, which is equivalent to $f36 = 6 + 6$, or $f36 = 12$. Thus, the value of $f36$ is 12.

▼ 1.24 — 40

The correct answer is 40. The y-intercept of a graph in the xy-plane is the point x, y on the graph where $x = 0$. The y-intercept of the graph shown is $0, 40$. Therefore, the value of y is 40.

▼ 1.25 — B

Choice B is correct. It's given that $hx = x^2 - 3$. Each table gives 1, 2, and 3 as the three given values of x. Substituting 1 for x in the equation $hx = x^2 - 3$ yields $h1 = 1^2 - 3$, or $h1 = -2$. Substituting 2 for x in the equation $hx = x^2 - 3$ yields $h2 = 2^2 - 3$, or $h2 = 1$. Finally, substituting 3 for x in the equation $hx = x^2 - 3$ yields $h3 = 3^2 - 3$, or $h3 = 6$. Therefore, hx is -2 when x is 1, hx is 1 when x is 2, and hx is 6 when x is 3. Choice B is a table with these values of x and their corresponding values of hx.

Choice A is incorrect. This is a table of values for the function $hx = x + 3$, not $hx = x^2 - 3$.

Choice C is incorrect. This is a table of values for the function $hx = 2x - 3$, not $hx = x^2 - 3$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.26 — A

Choice A is correct. The x-coordinate of any y-intercept of a graph is 0. Substituting 0 for x in the given equation yields $g0 = 11\frac{1}{12}^0$. Since any nonzero number raised to the 0th power is 1, this gives $g(0) = 11 \cdot 1$, or $g(0) = 11$. The y-intercept of the graph is, therefore, the point $0, 11$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.27 — A

Choice A is correct. It's given that the function $f(x) = 200{,}000(1.21)^x$ gives a company's predicted annual revenue, in dollars, x years after the company started selling light bulbs online. It follows that $f(x)$ represents the company's predicted annual revenue, in dollars, x years after the company started selling light bulbs online. Since the value of $f(5)$ is the value of $f(x)$ when $x = 5$, it follows that "$f(5)$ is approximately equal to $518{,}748$" means that $f(x)$ is approximately equal to $518{,}748$ when $x = 5$. Therefore, the best interpretation of the statement "$f(5)$ is approximately equal to $518{,}748$" in this context is 5 years after the company started selling light bulbs online, its predicted annual revenue is approximately $518{,}748$ dollars.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 1.28 — C

Choice C is correct. The value of $f(2)$ is the value of $f(x)$ when $x = 2$. Substituting 2 for x in the given function yields $f(2) = 2^3 + 15$, or $f(2) = 8 + 15$, which is equivalent to $f(2) = 23$. Therefore, the value of $f(2)$ is 23.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $f(2)$ when $f(x) = x3 + 15$, rather than $f(x) = x^3 + 15$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.29 — B

Choice B is correct. An x-intercept of a graph in the xy-plane is a point at which the graph crosses the x-axis. The graph shown crosses the x-axis at the point $(5, 0)$. Therefore, the x-intercept of the graph shown is $(5, 0)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.30 — A

Choice A is correct. It's given that the variable y represents the height, in meters, of the object above the ground. The graph shows that the height of the object was increasing from $x = 0$ to $x = 2$, and decreasing from $x = 2$ to $x = 4$. Therefore, the height of the object was increasing for the entire interval of time from $x = 0$ to $x = 2$.

Choice B is incorrect. The height of the object wasn't increasing for this entire interval of time, as it was decreasing from $x = 2$ to $x = 4$.

Choice C is incorrect. The height of the object was decreasing, not increasing, for this entire interval of time.

Choice D is incorrect. The height of the object was decreasing, not increasing, for this entire interval of time.

▼ 1.31 — C

Choice C is correct. It's given that the function g is defined by $gx = \sqrt{8x + 1}$. Substituting 3 for x in the given function yields $g3 = \sqrt{83 + 1}$, which is equivalent to $g3 = \sqrt{25}$, or $g3 = 5$. Therefore, the value of $g3$ is 5.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $83 + 1$, not $\sqrt{83 + 1}$.

▼ 1.32 — C

Choice C is correct. The vertex of the graph of a quadratic function in the xy-plane is the point at which the graph is either at its minimum or maximum y-value. In the graph shown, the minimum y-value occurs at the point $0, 2$.

Choice A is incorrect. The graph shown doesn't pass through the point $0, -2$.

Choice B is incorrect. The graph shown doesn't pass through the point $0, -3$.

Choice D is incorrect. The graph shown doesn't pass through the point $0, 3$.

▼ 1.33 — B

Choice B is correct. The value of $f144$ is the value of fx when $x = 144$. It's given that the function f is defined by $fx = 4 + \sqrt{x}$. Substituting 144 for x in this equation yields $f144 = 4 + \sqrt{144}$. Since the positive square root of 144 is 12, it follows that this equation can be rewritten as $f144 = 4 + 12$, or $f144 = 16$. Therefore, the value of $f144$ is 16.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f1{,}296$, not $f144$.

Choice D is incorrect. This is the value of $f5{,}184$, not $f144$.

▼ 1.34 — D

Choice D is correct. It's given that $fx = \frac{1}{6x}$. Substituting 3 for x in this equation yields $f3 = \frac{1}{63}$, or $f3 = \frac{1}{18}$. Therefore, when $x = 3$, the value of fx is $\frac{1}{18}$.

Choice A is incorrect. This is the value of fx when $x = 0.5$.

Choice B is incorrect. This is the value of fx when $x = 1$.

Choice C is incorrect. This is the value of fx when $x = 1.5$.

▼ 1.35 — 3

The correct answer is 3. For the graph of the exponential function f shown, where $y = fx$, it's given that the y-intercept of the graph is $0, y$. The graph intersects the y-axis at the point $0, 3$. Therefore, the value of y is 3.

▼ 1.36 — 31

The correct answer is 31. It's given that the y-intercept of the graph of $y = x^2 + 31$ in the xy-plane is $0, y$. Substituting 0 for x in the given equation yields $y = 0^2 + 31$, or $y = 31$. Thus, the value of y is 31.

▼ 1.37 — A

Choice A is correct. Substituting –1 for x in the equation that defines f gives $f(-1)=\frac{(-1)^2-6(-1)+3}{(-1)-1}$. Simplifying the expressions in the numerator and denominator yields $\frac{1+6+3}{-2}$, which is equal to $\frac{10}{-2}$ or –5.

Choices B, C, and D are incorrect and may result from misapplying the order of operations when substituting –1 for x.

▼ 1.38 — A

Choice A is correct. The *y*-intercept of a graph in the *xy*-plane is the point x,y on the graph where $x = 0$. For the graph shown, at $x = 0$, the corresponding value of y is -6. Therefore, the *y*-intercept of the graph shown is $0, -6$.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 1.39 — A

Choice A is correct. It's given that $fx = \frac{16}{x}$. Substituting 17 for x in this function yields $f17 = \frac{16}{17}$. Therefore, when $x = 17$, the value of fx is $\frac{16}{17}$.

Choice B is incorrect. This is the value of the reciprocal of fx when $x = 17$.

Choice C is incorrect. This is the value of fx when $x = 1$.

Choice D is incorrect. This is the value of x when $x = 17$.

▼ 1.40 — C

Choice C is correct. The value of $f(2)$ is found by evaluating the expression $2(3^x)$ when $x=2$. Substituting 2 for x in the given equation yields $f(2)=2(3^2)$. Simplifying 3^2 in the equation results in $f(2)=2(9)$. Evaluating the right-hand side of the equation yields $f(2)=18$. Therefore, the value of $f(2)$ is 18.

Choice A is incorrect and may result from evaluating the expression as (3^2). Choice B is incorrect and may result from evaluating the expression as $2(3 \cdot 2)$. Choice D is incorrect and may result from evaluating the expression as $(2 \cdot 3)^2$.

▼ 1.41 — C

Choice C is correct. It's given that $f(x) = x^3 + 9$. Substituting 2 for x in this equation yields $f(2) = 2^3 + 9$. This is equivalent to $f(2) = 8 + 9$, or $f(2) = 17$.

Choice A is incorrect. This is the value of $2 + 3 + 9$, not $2^3 + 9$.

Choice B is incorrect. This is the value of $23 + 9$, not $2^3 + 9$.

Choice D is incorrect. This is the value of $3^2 + 9$, not $2^3 + 9$.

▼ 1.42 — -3

The correct answer is -3. The y-intercept of the graph of a function in the xy-plane is the point where the graph crosses the y-axis. The graph of the polynomial function shown crosses the y-axis at the point $(0, -3)$. It's given that the y-intercept of the graph is $(0, y)$. Thus, the value of y is -3.

▼ 1.43 — D

Choice D is correct. If the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ down k units in the xy-plane, the function g can be defined by an equation of the form $g(x) = f(x) - k$. It's given that $f(x) = 7x^3$ and the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ down 2 units. Substituting $7x^3$ for $f(x)$ and 2 for k in the equation $g(x) = f(x) - k$ yields $g(x) = 7x^3 - 2$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This equation defines a function g for which the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ up, not down, 2 units.

▼ 1.44 — D

Choice D is correct. Because the graph of $y = fx$ is shown, the value of $f0$ is the value of y on the graph that corresponds with $x = 0$. When $x = 0$, the corresponding value of y is 3. Therefore, the value of $f0$ is 3.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 1.45 — B

Choice B is correct. The y-intercept of a graph in the xy-plane is the point at which the graph crosses the y-axis. The graph shown crosses the y-axis at the point $0, 2$. Therefore, the y-intercept of the graph shown is $0, 2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.46 — B

Choice B is correct. Each pair of values shown in the table gives the ordered pair of coordinates for a point that lies on the graph that represents the relationship between x and y in the xy-plane: $(0,0)$, $(1,1)$, $(2,8)$, and $(3,27)$. Only the graph in choice B passes through the points listed in the table that are visible in the given choices.

Choices A, C, and D are incorrect. None of these graphs passes through the point $(1,1)$.

▼ 1.47 — 23

The correct answer is 23. The value of $f2$ is the value of fx when $x = 2$. The function f is defined by $fx = x^3 + 15$. Substituting 2 for x in this equation yields $f2 = 2^3 + 15$. This equation can be rewritten as $f2 = 8 + 15$, or $f2 = 23$. Therefore, the value of $f2$ is 23.

▼ 1.48 — D

Choice D is correct. The value of $g4$ is the value of gx when $x = 4$. Substituting 4 for x in the given equation yields $g4 = 4 + 18$, which is equivalent to $g4 = 22$, or $g4 = 22$. Therefore, the value of $g4$ is 22.

Choice A is incorrect. This would be the value of $g4$ if function g was defined by $gx = -18$, not $gx = x + 18$.

Choice B is incorrect. This would be the value of $g4$ if function g was defined by $gx = -x$, not $gx = x + 18$.

Choice C is incorrect. This would be the value of $g4$ if function g was defined by $gx = -x + 18$, not $gx = x + 18$.

▼ 1.49 — 3

The correct answer is 3. A y-intercept of a graph in the xy-plane is a point x, y on the graph where $x = 0$. For the graph shown, at $x = 0$, the corresponding value of y is 3. Therefore, the y-coordinate of the y-intercept of the graph shown is 3.

▼ 1.50 — 77

The correct answer is 77. It's given that the function f is defined by $fx = x^2 + x + 71$. Substituting 2 for x in function f yields $f2 = 2^2 + 2 + 71$, which is equivalent to $f2 = 4 + 2 + 71$, or $f2 = 77$. Therefore, the value of $f2$ is 77.

▼ 1.51 — B

Choice B is correct. For the graph shown, the horizontal axis represents the number of years after a certain sport utility vehicle is first purchased, and the vertical axis represents the predicted value, in dollars, of the sport utility vehicle. According to the graph, 3 years after the sport utility vehicle is purchased, the predicted value of the sport utility vehicle is between $\$10{,}000$ and $\$15{,}000$. Of the given choices, only $\$13{,}632$ is between $\$10{,}000$ and $\$15{,}000$. Therefore, $\$13{,}632$ is closest to the predicted value of the sport utility vehicle 3 years after it is first purchased.

Choice A is incorrect. This is closest to the predicted value of the sport utility vehicle 5 years after it is first purchased.

Choice C is incorrect. This is closest to the predicted value of the sport utility vehicle 1 year after it is first purchased.

Choice D is incorrect. This is closest to the predicted value of the sport utility vehicle when it is first purchased.

▼ 1.52 — 7

The correct answer is 7. It's given that the x-intercept of the graph shown is $x, 0$. The graph passes through the point $7, 0$. Therefore, the value of x is 7.

▼ 1.53 — B

Choice B is correct. It's given that each day for five days after a social media channel posted a video, the number of subscribers doubled from the number the previous day. Since the number of subscribers doubled each day, the relationship between n and d can be represented by an exponential equation of the form $n = ab^d$, where a is the number of subscribers at the time of posting the video and the number of subscribers to the channel increases by a factor of b each day. It's given that at the time of posting the video, the channel had 53 subscribers. Therefore, $a = 53$. It's also given that the number of subscribers doubled, or increased by a factor of 2, from the number the previous day. Therefore, $b = 2$. Substituting 53 for a and 2 for b in the equation $n = ab^d$ yields $n = 532^d$.

Choice A is incorrect. This equation gives the total number of subscribers to a channel for which the initial number of subscribers was 1 and the number increased each day by 53 times the number from the previous day.

Choice C is incorrect. This equation gives the total number of subscribers to a channel for which the number of subscribers each day was half the number from the previous day, rather than double the number.

Choice D is incorrect and may result from conceptual errors.

▼ 1.54 — A

Choice A is correct. Since it's given that the account balance, At, in dollars, after t years can be modeled by an exponential function, it follows that function A can be written in the form $At = Nr^t$, where N is the initial value of the function and r is a constant related to the growth of the function. It's given that the initial balance of the account is $\$ 36{,}100.00$, so it follows that the initial value of the function, or N, must be $36{,}100.00$. Substituting $36{,}100.00$ for N in the equation $At = Nr^t$ yields $At = 36{,}100.00r^t$. It's given that the account balance after 13 years, or when $t = 13$, is $\$ 68{,}071.93$. It follows that $A13 = 68{,}071.93$, or $36{,}100.00r^{13} = 68{,}071.93$. Dividing each side of the equation $36{,}100.00r^{13} = 68{,}071.93$ by

$36{,}100.00$ yields $r^{13} = \frac{68{,}071.93}{36{,}100.00}$. Taking the 13th root of both sides of this equation yields $r = \sqrt[13]{\frac{68{,}071.93}{36{,}100.00}}$, or r is approximately equal to 1.05. Substituting 1.05 for r in the equation $At = 36{,}100.00r^t$ yields $At = 36{,}100.001.05^t$, so the equation $At = 36{,}100.001.05^t$ could define A.

Choice B is incorrect. Substituting 0 for t in this function indicates an initial balance of $\$\,31{,}971.93$, rather than $\$\,36{,}100.00$.

Choice C is incorrect. Substituting 0 for t in this function indicates an initial balance of $\$\,31{,}971.93$, rather than $\$\,36{,}100.00$. Additionally, this function indicates the account balance is decreasing, rather than increasing, over time.

Choice D is incorrect. This function indicates the account balance is decreasing, rather than increasing, over time.

▼ 1.55 — D

Choice D is correct. For the quadratic function $f(x) = x^2 + 4$, the vertex of the graph is $(0,4)$, and because the x^2 term is positive, the vertex is the minimum of the function. Choice D is the only option that meets these conditions.

Choices A, B, and C are incorrect. The vertex of each of these graphs doesn't correspond to the minimum of the given function.

▼ 1.56 — A

Choice A is correct. It's given that the function f gives a company's predicted annual revenue, in dollars, x years after the company started selling jewelry online. Since the value of $f(5)$ is the value of $f(x)$ when $x = 5$, it follows that "$f(5)$ is approximately equal to $648{,}650$" means that $f(x)$ is approximately equal to $648{,}650$ when $x = 5$. Therefore, the best interpretation of the given statement is that 5 years after the company started selling jewelry online, its predicted annual revenue is approximately $648{,}650$ dollars.

Choice B is incorrect. The function f gives the predicted annual revenue, not the predicted increase in annual revenue.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. In the given function, x represents the number of years after the company started selling jewelry online, not the percent increase in revenue from the previous year.

▼ 1.57 — 3

The correct answer is 3. The value of $f(0)$ is the value of y on the graph of $y = f(x)$ in the xy-plane that corresponds with $x = 0$. It's given that the value of $f(0)$ is an integer. For the graph of $y = f(x)$ shown, when $x = 0$, the corresponding integer value of y is 3. Therefore, the value of $f(0)$ is 3.

▼ 1.58 — D

Choice D is correct. It's given that the function f is defined by $fx = 5x^2$. Substituting 8 for x in $fx = 5x^2$ yields $f8 = 58^2$, which is equivalent to $f8 = 564$, or $f8 = 320$. Therefore, the value of $f8$ is 320.

Choice A is incorrect. This is the value of $f8$ if $fx = 5x$.

Choice B is incorrect. This is the value of $f8$ if $fx = 5x + 2$.

Choice C is incorrect. This is the value of $f8$ if $fx = 5x2$.

▼ 1.59 — A

Choice A is correct. When a graph is translated up 4 units, each point on the resulting graph is 4 units above the point on the original graph. In other words, the y-value of each point on the graph increases by 4. The graph shown passes through the points $1, -1$, $2, -2$, and $3, -1$. It follows that when the graph shown is translated up 4 units, the resulting graph will pass through the points $1, -1 + 4$, $2, -2 + 4$, and $3, -1 + 4$. These points are $1, 3$, $2, 2$, and $3, 3$, respectively. Of the given choices, only the graph in choice A passes through the points $1, 3$, $2, 2$, and $3, 3$.

Choice B is incorrect. This is the result of translating the graph down, rather than up, 4 units.

Choice C is incorrect. This is the result of translating the graph left, rather than up, 4 units.

Choice D is incorrect. This is the result of translating the graph right, rather than up, 4 units.

▼ 1.60 — D

Choice D is correct. The *x*-intercept of the graph shown is the point x, y on the graph where $y = 0$. At $y = 0$, the corresponding value of x is 4. Therefore, the *x*-intercept of the graph shown is $4, 0$.

Choice A is incorrect. This is the *x*-intercept of a graph in the *xy*-plane that intersects the *x*-axis at $x = -5$, not $x = 4$.

Choice B is incorrect. This is the *x*-intercept of a graph in the *xy*-plane that intersects the *x*-axis at $x = 5$, not $x = 4$.

Choice C is incorrect. This is the *x*-intercept of a graph in the *xy*-plane that intersects the *x*-axis at $x = -4$, not $x = 4$.

▼ 1.61 — C

Choice C is correct. Each point (x, y) on the graph represents the elevation x, in meters, and the corresponding measure of plant diversity y in a region of Switzerland. Therefore, the point on the graph with the greatest y-coordinate represents the location that has the greatest measure of plant diversity in the region. The greatest y-coordinate of any point on the graph is approximately 13,500. The x-coordinate of that point is approximately 1,250. Therefore, the closest elevation at which the plant diversity is the greatest is 1,250 meters.

Choice A is incorrect. This value is closest to the greatest y-coordinate of any point on the graph and therefore represents the greatest measure of plant diversity, not the elevation where the greatest measure of plant diversity occurs. Choice B is incorrect. At an elevation of 3,000 meters the measure of plant diversity is approximately 4,000. Because there are points on the graph with greater y-coordinates, 4,000 can't be the greatest measure of plant diversity, and 3,000 meters isn't the elevation at which the greatest measure of plant diversity occurs. Choice D is incorrect. At an elevation of 250 meters, the measure of plant diversity is approximately 11,000. Because there are points on the graph with greater y-coordinates, 11,000 can't be the greatest measure of plant diversity and 250 meters isn't the elevation at which the greatest measure of plant diversity occurs.

▼ 1.62 — C

Choice C is correct. Adding the like terms x and $0.25x$ yields the equation $f(x) = (1.25x)(50 - x)$. Substituting 20 for x yields $f(20) = (1.25(20))(50 - 20)$. The product

$1.25(20)$ is equal to 25, and the difference $50-20$ is equal to 30. Substituting these values in the given equation gives $f(20)=(25)(30)$, and multiplying 25 by 30 results in $f(20)=750$.

Choices A, B, and D are incorrect and may result from conceptual or computational errors when finding the value of $f(20)$.

▼ 1.63 — D

Choice D is correct. The *y*-intercept of a graph in the *xy*-plane is the point at which the graph crosses the *y*-axis. The graph shown crosses the *y*-axis at the point $0, 8$. Therefore, the *y*-intercept of the graph shown is $0, 8$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.64 — A

Choice A is correct. It's given that in the *xy*-plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ up 4 units. It follows that $g(x) = f(x) + 4$. It's also given that function *f* is defined by $f(x) = 3x^5$. Substituting $3x^5$ for $f(x)$ into $g(x) = f(x) + 4$ yields $g(x) = 3x^5 + 4$. Therefore, the equation that defines function *g* is $g(x) = 3x^5 + 4$.

Choice B is incorrect. This equation defines a function *g* for which the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ down, not up, 4 units.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 1.65 — C

Choice C is correct. It's given that one of the two players in a match is eliminated during each round. Therefore, the number of players decreases by half at the end of each round, and so the number of players eliminated at the end of a round also decreases by half. A decreasing exponential equation can be written in the form $p = a(1 - b)^r$, where *b* is the value of the decrease each round, *r*, and *a* is the value of *p*, the number of players, when $r = 0$.

It's given that there are 240 players in the tennis competition. Therefore, $a = 240$. Since the number of players eliminated at the end of a round decreases by half, it follows that $b = \frac{1}{2}$. Substituting 240 for a and $\frac{1}{2}$ for b in the equation $p = a(1 - b)^r$ yields $p = 240(1 - \frac{1}{2})^r$, or $p = 240(\frac{1}{2})^r$.

Choice A is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which there are 15, not 240, players.

Choice B is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which there are 15, not 240, players, and for which the number of players doubles, rather than decreases by half, after each round.

Choice D is incorrect. This equation gives the number of players eliminated at the end of round r for a competition in which the number of players doubles, rather than decreases by half, after each round.

▼ 1.66 — 15000

The correct answer is $15{,}000$. Let y represent the population of Smithville x years after 1659. Since the population doubled every 75 years, the relationship between x and y can be represented by an exponential equation in the form $y = ab^{\frac{x}{k}}$, where a is the population of Smithville in 1659 and the population increases by a factor of b every k years. It's given that the population doubled, or increased by a factor of 2, every 75 years. Therefore, $b = 2$ and $k = 75$. Substituting 2 for b and 75 for k in the equation $y = ab^{\frac{x}{k}}$ yields $y = a2^{\frac{x}{75}}$. It's also given that the population of this town was $240{,}000$ in 1959. The number of years from 1659 to 1959 is $1959 - 1659$, or 300 years. Therefore, when $x = 300$, the value of y is $240{,}000$. Substituting 300 for x and $240{,}000$ for y in the equation $y = a2^{\frac{x}{75}}$ yields $240{,}000 = a2^{\frac{300}{75}}$, or $240{,}000 = a2^4$, which is equivalent to $240{,}000 = a(16)$. Dividing both sides of this equation by 16 yields $15{,}000 = a$. Therefore, the population of this town in 1659 was $15{,}000$.

▼ 1.67 — A

Choice A is correct. It's given that $pn = 7n^3$. Substituting 56 for pn in this equation yields $56 = 7n^3$. Dividing each side of this equation by 7 yields $8 = n^3$. Taking the cube root of each side of this equation yields $2 = n$. Therefore, when pn is equal to 56, the value of n is 2.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.68 — C

Choice C is correct. It's given that the function f is defined by $fx = x - 4x$. It's also given that $f5 - fa = -15$. Substituting 5 for x in the function $fx = x - 4x$ yields $f5 = 5 - 45$ and substituting a for x in the function $fx = x - 4x$ yields $fa = a - 4a$. Therefore, $f5 = 15$ and $fa = -3a$. Substituting 15 for $f5$ and $-3a$ for fa in the equation $f5 - fa = -15$ yields $15 - -3a = -15$. Subtracting 15 from both sides of this equation yields $--3a = -30$. Dividing both sides of this equation by -1 yields $-3a = 30$. By the definition of absolute value, if $-3a = 30$, then $-3a = 30$ or $-3a = -30$. Dividing both sides of each of these equations by -3 yields $a = -10$ or $a = 10$, respectively. Thus, of the given choices, a value of a that satisfies $f5 - fa = -15$ is 10.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.69 — 3600

The correct answer is $3{,}600$. The predicted number of bacteria initially measured in the growth medium is the value of y when $x = 0$. Substituting 0 for x in the given equation yields $y = 3{,}600(a)^0$. Since a is positive, this equation is equivalent to $y = 3{,}600(1)$. Therefore, when $x = 0$, the value of y is $3{,}600$. Thus, the predicted number of bacteria initially measured in the growth medium is $3{,}600$.

▼ 1.70 — A

Choice A is correct. For an exponential function in the form $fx = ab^x$, where a and b are positive constants and $b < 1$, the initial value of $f(x)$, or the value of $f(x)$ when $x = 0$, is a and the value of $f(x)$ decreases by $100(1 - b)\%$ each time x increases by 1. Therefore, the initial value of the function $fx = 4{,}0000.75^x$, or the value of $f(x)$ when $x = 0$, is $4{,}000$. Therefore, the best interpretation of $4{,}000$ in this context is the estimated initial beetle population for this species and area in 2012.

Choice B is incorrect. The estimated beetle population for this species and area 7 years after 2012 is $4{,}000(0.75)^7$, or approximately 534, not $4{,}000$.

Choice C is incorrect. The estimated percent decrease in the beetle population for this species and area each year after 2012 is $100(1 - 0.75)$, or 25, not $4{,}000$.

Choice D is incorrect. The estimated percent decrease in the beetle population for this species and area every 7 years after 2012 is $100(1 - 0.75^{7})$, or approximately 87, not $4{,}000$.

▼ 1.71 — B

Choice B is correct. It's given that in January 2018, there were $1{,}300$ customers subscribed to a company's newsletter and for the next 24 months after January 2018, the total number of customers subscribed to the newsletter each month was 7% greater than the total number subscribed the previous month. It follows that this situation can be represented by the equation $c = a1 + \frac{r}{100}^{m}$, where c is the total number of customers subscribed to the company's newsletter m months after January 2018, a is the number of customers subscribed to the newsletter in January 2018, and the total number of customers subscribed to the newsletter each month was $r\%$ greater than the total number subscribed the previous month. Substituting $1{,}300$ for a and 7 for r in this equation yields $c = 1{,}3001 + \frac{7}{100}^{m}$, or $c = 1{,}3001.07^{m}$.

Choice A is incorrect. This equation represents a situation where the total number of customers subscribed each month was 93% less, not 7% greater, than the total number subscribed the previous month.

Choice C is incorrect. This equation represents a situation where the total number of customers subscribed each month was 70%, not 7%, greater than the total number subscribed the previous month.

Choice D is incorrect. This equation represents a situation where the total number of customers subscribed each month was 600%, not 7%, greater than the total number subscribed the previous month.

▼ 1.72 — D

Choice D is correct. The value of $f0$ is the value of fx when $x = 0$. Substituting 0 for x in the given function yields $f0 = 2700.1^{0}$, or $f0 = 2701$, which is equivalent to $f0 = 270$. Therefore, the value of $f0$ is 270.

Choice A is incorrect. This is the value of x, not fx.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f1$, not $f0$.

▼ 1.73 — C

Choice C is correct. It's given that f is a polynomial function and the graph of $y = f(x)$ in the xy-plane passes through the points $(-5, 0)$, $(1, 0)$, and $(4, 0)$. A point in the form $(a, 0)$ on the graph of $f(x)$ means that $x - a$ is a factor of $f(x)$. Since the graph of the function f passes through the point $(1, 0)$, $x - 1$ must be a factor of $f(x)$.

Choice A is incorrect. This is a factor if the graph of $y = f(x)$ passed through $(-1, 0)$.

Choice B is incorrect. This is a factor if the graph of $y = f(x)$ passed through $(-4, 0)$.

Choice D is incorrect. This is a factor if the graph of $y = f(x)$ passed through $(5, 0)$.

▼ 1.74 — A

Choice A is correct. The variable t represents the seconds after the object is launched. Since $h(0) = 72$, this means that the height, in feet, at 0 seconds, or the initial height, is 72 feet.

Choices B, C, and D are incorrect and may be the result of misinterpreting the function in context.

▼ 1.75 — C

Choice C is correct. This exponential growth model can be written in the form $p(t) = A(1 + r)^t$, where $p(t)$ is the population t years after 1788, A is the initial population, and r is the yearly growth rate, expressed as a decimal. Since there were 24 jackrabbits in Australia in 1788, $A = 24$. Since the number of jackrabbits increased by an average of 16.2% each year, $r = 0.162$. Therefore, the equation that best models this situation is $p(t) = 24(1.162)^t$.

Choices A, B, and D are incorrect and may result from misinterpreting the form of an exponential growth model.

▼ 1.76 — B

Choice B is correct. It's given that $fx = (x + 6)(x + 5)(x - 4)$ and $y = f(x) - 3$. Substituting $(x + 6)(x + 5)(x - 4)$ for fx in the equation $y = f(x) - 3$ yields $y = (x + 6)(x + 5)(x - 4) - 3$. Substituting -6 for x in this equation yields $y = (-6 + 6)(-6 + 5)(-6 - 4) - 3$, or $y = -3$. Substituting -5 for x in the equation $y = (x + 6)(x + 5)(x - 4) - 3$ yields

$y = (-5 + 6)(-5 + 5)(-5 - 4) - 3$, or $y = -3$. Substituting 4 for x in the equation $y = (x + 6)(x + 5)(x - 4) - 3$ yields $y = (4 + 6)(4 + 5)(4 - 4) - 3$, or $y = -3$. Therefore, when $x = -6$ then $y = -3$, when $x = -5$ then $y = -3$, and when $x = 4$ then $y = -3$. Thus, the table of values in choice B represents $y = f(x) - 3$.

Choice A is incorrect. This table represents $y = x - 3$ rather than $y = f(x) - 3$.

Choice C is incorrect. This table represents $y = x + 3$ rather than $y = f(x) - 3$.

Choice D is incorrect. This table represents $y = fx + 3$ rather than $y = f(x) - 3$.

▼ 1.77 — C

Choice C is correct. It's given that a function p estimates that there were $2{,}000$ animals in a population in 1998 and that each year from 1998 to 2010, the number of animals in this population increased by 3% of the number of animals in the population the previous year. It follows that this situation can be represented by the function $px = a1 + \frac{r}{100}^{x}$, where px is the estimated number of animals in the population x years after 1998, a is the estimated number of animals in the population in 1998, and each year the estimated number of animals increased by $r\%$. Substituting $2{,}000$ for a and 3 for r in this function yields $px = 2{,}0001 + \frac{3}{100}^{x}$, or $px = 2{,}0001.03^{x}$.

Choice A is incorrect. This function represents a population in which each year the number of animals increased by 200%, not 3%, of the number of animals in the population the previous year.

Choice B is incorrect. This function represents a population in which each year the number of animals increased by 97%, not 3%, of the number of animals in the population the previous year.

Choice D is incorrect. This function represents a population in which each year the number of animals decreased, rather than increased, by 3% of the number of animals in the population the previous year.

▼ 1.78 — D

Choice D is correct. Since the number of bacteria doubles every day, the relationship between t and y can be represented by an exponential equation of the form $y = ab^{t}$, where a is the number of bacteria at the start of the observation and the number of bacteria increases by a factor of b every day. It's given that there are $44{,}000$ bacteria at the start of

the observation. Therefore, $a = 44{,}000$. It's also given that the number of bacteria doubles, or increases by a factor of 2, every day. Therefore, $b = 2$. Substituting $44{,}000$ for a and 2 for b in the equation $y = ab^t$ yields $y = 44{,}0002^t$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation represents a situation where the number of bacteria is decreasing by half, not doubling, every day.

▼ 1.79 — B

Choice B is correct. It's given that the graph shows the estimated boiling point y, in degrees Celsius, of a normal paraffin with a molecular weight of x grams per mole. It follows that for the point $(149.02, 186.05)$, 149.02 represents the molecular weight, in grams per mole, of a normal paraffin and 186.05 represents its estimated boiling point, in degrees Celsius. Therefore, the best interpretation of the point $(149.02, 186.05)$ is that a normal paraffin with a molecular weight of 149.02 grams per mole has an estimated boiling point of 186.05 degrees Celsius.

Choice A is incorrect. This statement represents a situation where x, not y, represents the boiling point, in degrees Celsius, and y, not x, represents the molecular weight, in grams, per mole.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

▼ 1.80 — D

Choice D is correct. Let y represent the number of cells per milliliter x hours after the initial observation. Since the number of cells per milliliter doubles every 3 hours, the relationship between x and y can be represented by an exponential equation of the form $y = ab^{\frac{x}{k}}$, where a is the number of cells per milliliter during the initial observation and the number of cells per milliliter increases by a factor of b every k hours. It's given that there were $300{,}000$ cells per milliliter during the initial observation. Therefore, $a = 300{,}000$. It's also given that the number of cells per milliliter doubles, or increases by a factor of 2, every 3 hours. Therefore, $b = 2$ and $k = 3$. Substituting $300{,}000$ for a, 2 for b, and 3 for k in the equation $y = ab^{\frac{x}{k}}$ yields $y = 300{,}0002^{\frac{x}{3}}$. The number of cells per milliliter there will be 15 hours after the initial

observation is the value of y in this equation when $x = 15$. Substituting 15 for x in the equation $y = 300{,}000 2^{\frac{x}{3}}$ yields $y = 300{,}000 2^{\frac{15}{3}}$, or $y = 300{,}000 2^{5}$. This is equivalent to $y = 300{,}000 \cdot 32$, or $y = 9{,}600{,}000$. Therefore, 15 hours after the initial observation, there will be $9{,}600{,}000$ cells per milliliter.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.81 — 14, -5, -4

The correct answer is either 14, -5, or -4. The x-intercepts of a graph in the xy-plane are the points at which the graph intersects the x-axis, or when the value of y is 0. Substituting 0 for y in the given equation yields $0 = 3x - 14x + 5x + 4$. Dividing both sides of this equation by 3 yields $0 = x - 14x + 5x + 4$. Applying the zero product property to this equation yields three equations: $x - 14 = 0$, $x + 5 = 0$, and $x + 4 = 0$. Adding 14 to both sides of the equation $x - 14 = 0$ yields $x = 14$, subtracting 5 from both sides of the equation $x + 5 = 0$ yields $x = -5$, and subtracting 4 from both sides of the equation $x + 4 = 0$ yields $x = -4$. Therefore, the x-coordinates of the x-intercepts of the graph of the given equation are 14, -5, and -4. Note that 14, -5, and -4 are examples of ways to enter a correct answer.

▼ 1.82 — B

Choice B is correct. It's given that the graph shows the height above the water y, in meters, of a diver x seconds after diving from a platform. The x-intercept of a graph is the point at which the graph intersects the x-axis, or when the value of y is 0. The graph shown intersects the x-axis between $x = 1$ and $x = 2$. In other words, the diver is 0 meters above the water, or hits the water, between 1 and 2 seconds after diving from the platform. Of the given choices, only choice B includes an interpretation where the diver hits the water between 1 and 2 seconds. Therefore, the best interpretation of the x-intercept of the graph is the diver hits the water at 1.6 seconds.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the best interpretation of the maximum value, not the x-intercept, of the graph.

Choice D is incorrect and may result from conceptual errors.

▼ 1.83 — A

Choice A is correct. In the given function, st represents the approximate mass, in grams, of the sample that remains t years after the sample starts to decay. It follows that the best interpretation of $s87 = 23$ is that approximately 23 grams of the sample remains 87 years after the sample starts to decay.

Choice B is incorrect. The mass of the sample has decreased by approximately $184 - 23$, or 161, grams, not 23 grams, 87 years after the sample starts to decay.

Choice C is incorrect. The mass of the sample has decreased by approximately 78 grams, not 87 grams, 23 years after the sample starts to decay.

Choice D is incorrect. This would be the best interpretation of $s23 = 87$, not $s87 = 23$.

▼ 1.84 — -6, 4

The correct answer is either -6 or 4. The x-intercepts of a graph in the xy-plane are the points where $y = 0$. Thus, for an x-intercept of the graph of $y = fx$, $0 = fx$. Substituting 0 for fx in the equation $fx = x + 6x - 4$ yields $0 = x + 6x - 4$. By the zero product property, $x + 6 = 0$ and $x - 4 = 0$. Subtracting 6 from both sides of the equation $x + 6 = 0$ yields $x = -6$. Adding 4 to both sides of the equation $x - 4 = 0$ yields $x = 4$. Therefore, the x-coordinates of the x-intercepts of the graph of $y = fx$ are -6 and 4. Note that -6 and 4 are examples of ways to enter a correct answer.

▼ 1.85 — C

Choice C is correct. It's given that the relationship between t and n is exponential. The table shows that the value of n increases as the value of t increases. Therefore, the relationship between t and n can be represented by an increasing exponential equation of the form $n = a1 + b^t$, where a and b are positive constants. The table shows that when $t = 0$, $n = 604$. Substituting 0 for t and 604 for n in the equation $n = a1 + b^t$ yields $604 = a1 + b^0$, which is equivalent to $604 = a1$, or $604 = a$. Substituting 604 for a in the equation $n = a1 + b^t$ yields $n = 6041 + b^t$. The table also shows that when $t = 1$, $n = 606.42$. Substituting 1 for t and 606.42 for n in the equation $n = 6041 + b^t$ yields $606.42 = 6041 + b^1$, or $606.42 = 6041 + b$. Dividing both sides of this equation by 604 yields approximately $1.004 = 1 + b$. Subtracting 1 from both sides of this equation yields that the value of b is approximately 0.004. Substituting 0.004 for b in the equation $n = 6041 + b^t$ yields $n = 6041 + 0.004^t$. Therefore, of the choices, choice C best represents the relationship between t and n.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.86 — B

Choice B is correct. In the statement "$m(330)$ is approximately equal to 362," the input of the function, 330, is the value of t, the elapsed time, in days, since the animal was born. The approximate value of the function, 362, is the predicted body mass, in kilograms, of the animal after that time has elapsed. Therefore, the predicted body mass of the animal was approximately 362 kg 330 days after it was born.

Choice A is incorrect. This would be the best interpretation of the statement "$m(362)$ is approximately equal to 330."

Choice C is incorrect. The number $\frac{330}{7}$ is the number of weeks, not the number of days, after the animal was born.

Choice D is incorrect. This would be the best interpretation of the statement "$m(362)$ is approximately equal to $\frac{330}{7}$."